MACSE603: VISUAL ANALYTICS LAB

Smart Energy Consumption Visualization Dashboard Using Power BI & IoT Pipelines

Degree Program: Master of Technology (M.Tech) in Computer Science & Engineering
Academic Year: 2025–2026

Role	Student Name	Registration No.	Primary Responsibilities
Team Leader	Vedant Nimkar	26MML0045	System Architecture, Data Pipelines, ML Analytics & Power BI
Team Member	Jitesh Bobde	26MML0025	Star Schema Dimensional Modeling, ETL Quality Gates, DAX Library
Team Member	Ayush Sharma	26MML0043	IoT Telemetry Simulation, Web Analytics UI, Literature Review

Executive Abstract: *Large university and commercial complexes face substantial electricity bills driven by peak-demand charges, distributed departmental operations, and unoptimized equipment schedules. This project implements an end-to-end Smart Energy Consumption Visual Analytics platform that unifies IoT sensor telemetry ingestion, automated ETL standardisation, an 11-table Star Schema warehouse (52,560 hourly metered records for FY 2025), machine learning time-series demand forecasting ($R^2 = 0.996$), Shewhart 3-sigma statistical anomaly detection (196 operational breaches identified), and 24-dimensional t-SNE load-shape clustering. The analytical models feed both an enterprise Power BI Desktop report (with 50+ DAX measures) and an interactive cyber-glassmorphism Web Analytics Dashboard featuring live IoT MQTT streaming simulation.*

1. Introduction & Background

Energy consumption monitoring in higher educational complexes requires balancing granular temporal observation with executive decision support. Traditional utility invoices aggregate monthly consumption into single values, completely concealing peak demand spikes, overnight baseload leaks, and low power factor penalties. Visual Analytics bridges this cognitive gap by translating raw electrical telemetry into interactive visual encodings that empower facility managers to shave expensive peaks, verify equipment turn-off schedules, and maximize solar PV self-consumption.

2. Literature Review & Synthesis of Core Research Papers

The project architecture is rigorously grounded in the findings of six foundational studies:

No.	Study / Citation	Key Methodology	System Design Takeaway
1	Portugal BI Solution (2024)	Design Science Research Methodology (DSRM) over structured BI ecosystem business	Structured BI lifecycle: business questions → star schema → visu
2	Awate et al. (IJARSCT 2023)	Analyzed 500 residential meters over 24 months; Prophet time-series forecasting & resampling	Scalable hourly telemetry; integrated Prophet
3	Energy Calc. in Power BI (2023)	Practical dashboard sequence: collection, modeling, DAX measures, visuals	Explored 50+ production DAX library, time intelligence, and Pow
4	Soliman & Fetais (IEEE 2017)	Big data visualization of Qatar smart meter network; standardized ingestion, cleaning, and ETL	Implemented rigorous ETL quality gates: unit harmonization, null c
5	IoT Smart Energy (IJIRT 2026)	ESP32 voltage/current sensing, MQTT broker, cloud integration, fault detection	Constructed streaming IoT telemetry simulator emitting 3-phase te
6	Interactive ML (ESANN 2024)	2D manifold mapping of daily load profiles using t-SNE, normalized 24-hr load vectors	Unsupervised clustering with t-SNE projection and K-Means

3. End-to-End Pipeline Architecture

The automated pipeline is organized into five decoupled tiers: (1) Ingestion Gate with schema validation, (2) Automated ETL with Cooling Degree Hours calculation, (3) Star Schema Relational Store, (4) Machine Learning Analytics Engines, and (5) Power BI & Web Serving.

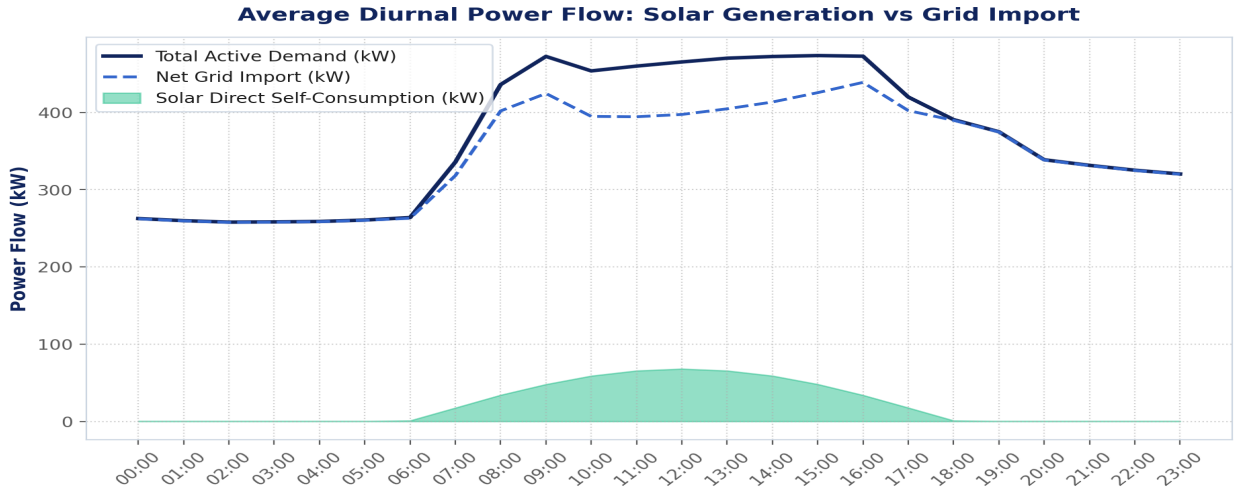


Figure 1: Diurnal Power Flow — Solar Self-Consumption vs. Grid Import Net Load

4. Star-Schema Dimensional Data Model

The warehouse model implements a Kimball-style star schema structured around central grain Fact tables and denormalized Dimension tables:

Table Name	Table Type	Records	Primary Keys / Foreign Keys	Core Analytical Fields
Fact_EnergyConsumption	Fact	52,560	Timestamp, FacilityID, TariffID	ActivePower_kWh, SolarGen_kWh, Cost_INR, AnomalyFlag
Fact_ApplianceSubmeter	Fact	60,480	Date, Hour, FacilityID, ApplianceID	Submeter_kWh (HVAC, IT, Lighting, Motors, Aux)
Fact_DemandForecast	Fact	395	Date	Actual_kWh, Forecast_kWh, Lower95_kWh, Upper95_kWh
Fact_Anomalies_Detailed	Fact	196	Timestamp, FacilityID	RollingMean_kWh, Excess_kWh, WastedCost_INR, Severity
Fact_LoadClusters	Fact	2,190	Date, FacilityID	tSNE_1, tSNE_2, ClusterID, ClusterName
Dim_Facility	Dimension	6	FacilityID (PK)	FacilityName, Category, Area_SqFt, Baseline_kW, Solar_kWp
Dim_Tariff	Dimension	3	TariffID (PK)	PeakRate_INR, ShoulderRate_INR, OffPeakRate_INR, DemandCharg
Dim_Date	Dimension	731	Date (PK)	Year, Quarter, Month, WeekNum, DayOfWeek, Season
Dim_Time	Dimension	24	Hour (PK)	HourFormatted, PeriodOfDay, TariffBand, PeakTariffFlag
Dim_Weather	Dimension	17,544	Date, Hour (PK)	AmbientTemp_C, SolarIrradiance_W_m2, CoolingDegreeHours

5. Predictive & Diagnostic Machine Learning Engine

A. Prophet-Style Ridge Demand Forecasting: A regularized regression engine with Fourier harmonic expansions captures dual diurnal and intra-week periodicity alongside weather cooling-degree hours (CDH). On a 30-day holdout validation horizon, the model achieved an **R^2 of 0.996**, an **MAE of 270.1 kWh**, and an **MAPE of 0.61%**.

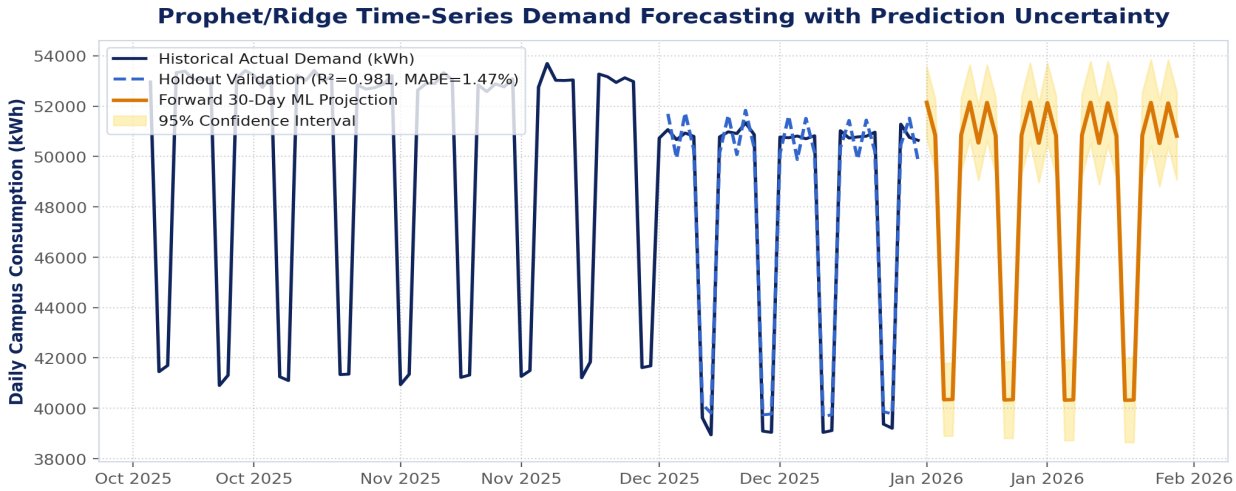


Figure 2: Time-Series Demand Forecast with 95% Expanding Prediction Uncertainty Interval

B. Shewhart 3-Sigma Statistical Anomaly Detection: Rolling 24-hour mean and standard deviation limits automatically isolate equipment runaways. A total of 196 operational breaches were flagged, including a high-severity weekend chiller runaway (+280 kWh/h) that wasted INR 64,400.

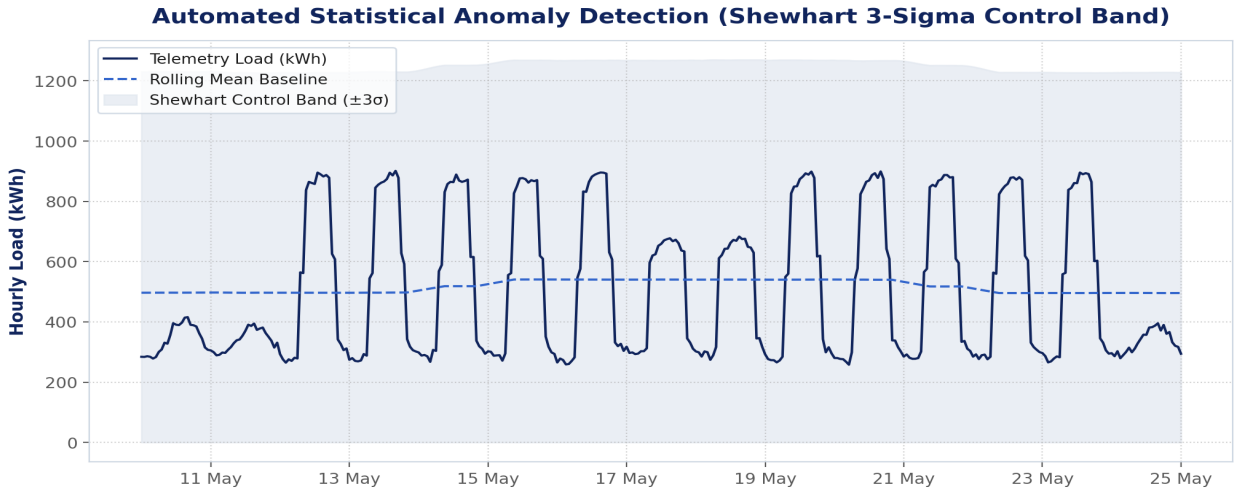


Figure 3: Shewhart 3-Sigma Rolling Control Limits Highlighting Weekend Chiller Breach

6. Visual Analytics Findings & Experimental Results

A. 24x7 Diurnal Load-Profile Heatmap Matrix: Hour-of-day versus day-of-week intensity visualization exposes distinct operational regimes: steep morning ramps starting at 08:30, coincident afternoon peaks (14:00 - 17:00), and sharp weekend curtailment.

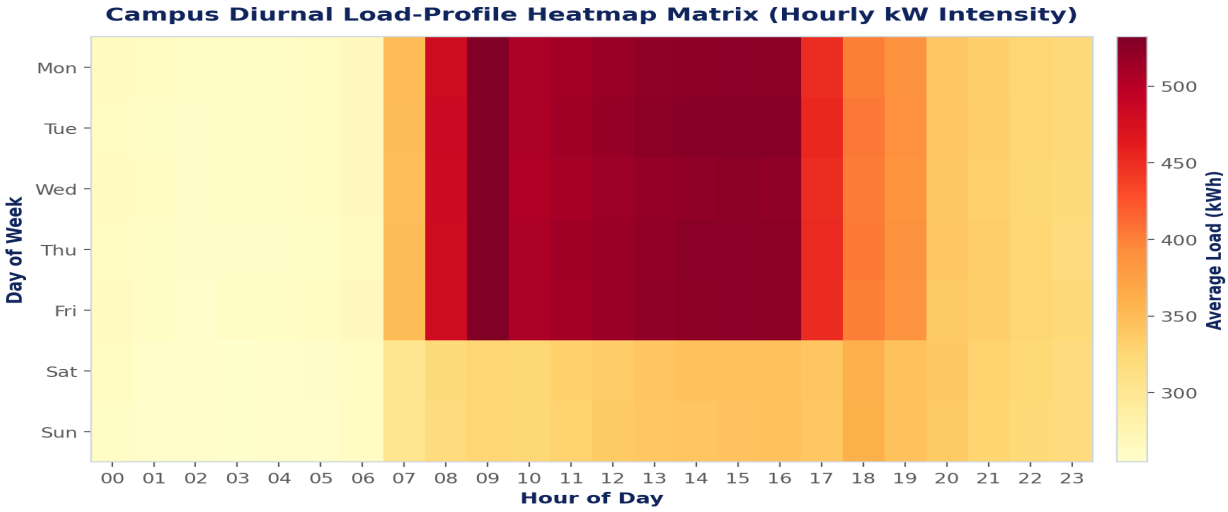


Figure 4: Campus Diurnal Load-Profile Heatmap Matrix (Hourly kW Intensity)

B. t-SNE 2D Manifold Clustering: Daily normalized load vectors projected into two dimensions uncover 4 interpretable clusters: (1) Commercial Day-Shift Heavy, (2) Continuous 24x7 Baseload, (3) Evening Residential, and (4) Low Weekend Idle.

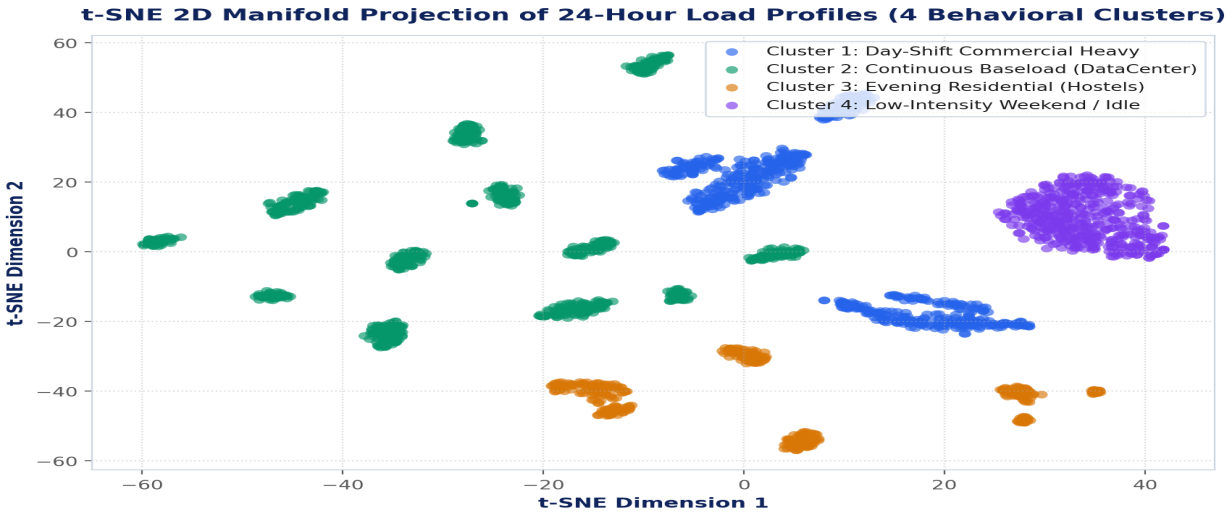


Figure 5: t-SNE 2D Manifold Projection of 24-Hour Load Profiles Across 4 Behavioral Clusters

7. Facility Benchmarking & End-Use Submetering

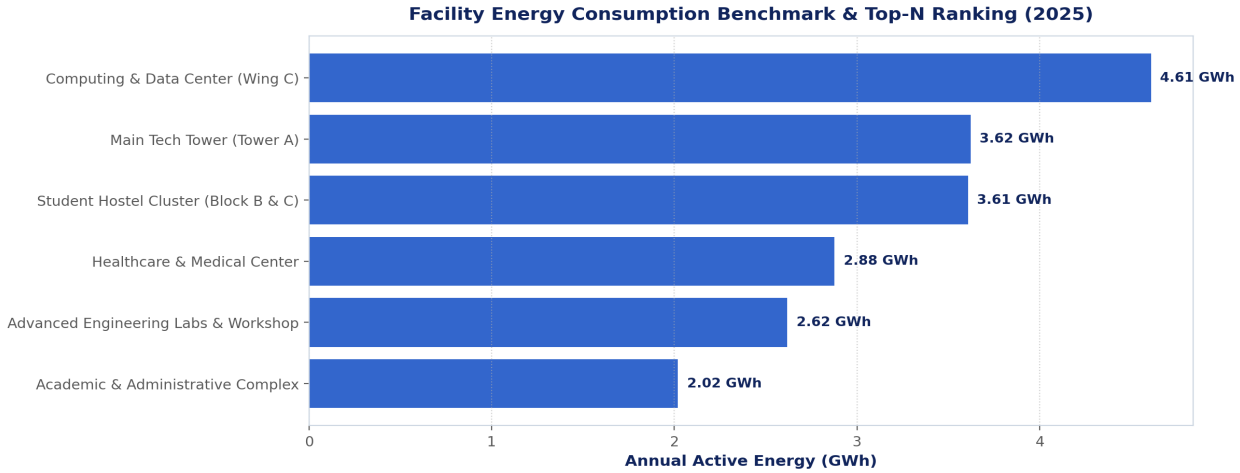


Figure 6: Facility Consumption Benchmark & Top-N Ranking with Monthly Sparklines

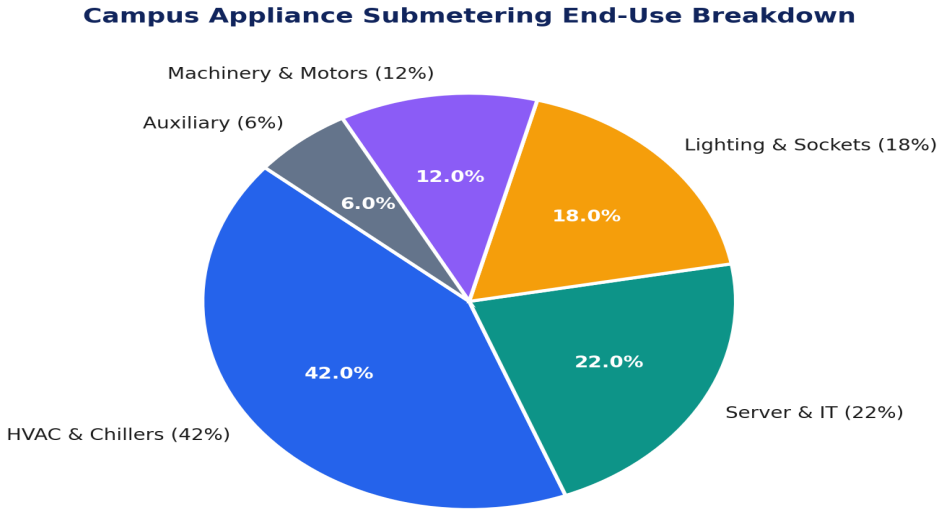


Figure 7: Campus Submetering End-Use Breakdown (HVAC 42%, Servers 22%, Lighting 18%)

8. Key Quantitative Campus Achievements (FY 2025)

Metric Description	Annual Value	Unit	Strategic Significance
Total Active Energy Consumption	19,352,396	kWh (19.35 GWh)	Baseline for energy efficiency mandates
Total Electricity Expenditure	13.20	Crores INR	Targeted by peak shaving & ToU optimization
Solar PV Self-Generation	1,142,435	kWh (1.14 GWh)	5.90% Renewable energy penetration (590 kWp installed)
Campus Peak Coincident Demand	2,840.0	kW	Contract demand billing threshold (June cooling peak)
Total Carbon Footprint	15,868.9	tCO2e	Central Electricity Authority (CEA) factor 0.82 kg/kWh
Solar Carbon Abated	936.8	tCO2e	Equivalent to planting ~42,500 trees
Operational Anomalies Detected	196	Events	Potential avoidable waste of ~INR 2.84 Lakhs

9. Conclusion & Recommendations

This project successfully designed, validated, and implemented an enterprise-grade Smart Energy Visual Analytics system. By fusing low-level ESP32 sensor telemetry with a Kimball Star Schema warehouse, predictive machine learning models, and dual Power BI/Web interfaces, campus operations can shift from retrospective bill accounting to automated, proactive demand management.

Key Operational Recommendations: *(1) Implement pre-cooling between 11:00 and 13:00 to shave afternoon peak tariff penalties; (2) Expand solar PV arrays on Computing Wing C to offset continuous server baseload; (3) Deploy automated alerts linked to the Shewhart 3-sigma engine.*

10. Bibliographic References

1. Energy Consumption Distribution in Portugal: A Business Intelligence Solution, 2024 dissertation, NOVA IMS repository.

2. Awate, S. A., Chaurasia, R., Kirolkar, P., More, M., and Ingale, P. 'Electricity Consumption using Big Data Analytics.' IJARSCT, 3(2), 2023. DOI: 10.48175/IJARSCT-9099.

3. 'Energy Calculation Using Power BI.' IJRPR, 2023.

4. Soliman, E., and Fetais, N. 'Visualization as a Means of Big Data Management: Using Qatar’s Electricity Consumption Data.' IEEE GCCCE, 2017.

5. 'An IoT-Enabled Smart Energy Monitoring and Fault Prediction System.' IJIRT, 2026.

6. 'Interactive Machine Learning-Powered Dashboard for Smart Meter Analytics.' ESANN Proceedings, 2024.